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COLLEGE OF ENGINEERING
A U T O N O M O U S

PROJECT TITLE:

MEDISENSE: AN AI-DRIVEN
PATIENT MEDICATION NON-
ADHERENCE AND CHRONIC
HEALTH RISK MITIGATION
PORTAL

24MCAR295 - Mini Project

Scrum Master

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ABSTRACT

MediSense: AI-Driven Patient Medication Non-Adherence and Chronic Health Risk Mitigation Portal

MediSense is a proactive, intelligent, healthcare portal designed to monitor, analyze, and predict medication non-adherence of patients suffering from chronic diseases like diabetes, hypertension, and cardiovascular diseases. The platform continuously monitors patient behavioural patterns such as medicine refill frequencies, daily vital logs, etc. The system uses these data to predict the probability of the patient neglecting the treatment. If a patient starts skipping doses, delaying pharmacy pickups, or cancelling scheduled doctor appointments, the backend automatically observes these deviations. If this behavior crosses a dangerous threshold, the system immediately alerts the concerned medical professionals for taking necessary actions. This way the clinicians can easily identify the high-risk patients and provide them more care and medical aid. This system is built using **HTMX, Tailwind CSS, Alpine.js** and **Chart.js** as frontend and **Python Django + Fast API** as backend.

Key Features and User Roles

1. Patient

- **Account Management:** Register, login, and manage personal profiles.
- **Pill Checklist:** Update the daily status of consuming medicines exactly as prescribed by the doctor.
- **Daily Vitals Input:** Update clinical parameters like Blood Pressure(BP) and Blood Sugar levels directly into their profile.
- **View Medicines and Prescription :** View the complete active prescription history, digital charts, and its upcoming refill dates issued by the clinic.



2. Doctor/ Medical Professional

- **Account Management:** Register, login, and manage personal profiles.
- **AI Risk Monitor:** A real-time predictive dashboard that highlights patients who are non-adherent to their medicines.

3. Admin

- **User Management:** Oversee patient and doctor registrations, manage user accounts, and resolve any issues.
- **Pharmacy Verification:** Manually update the exact status patient medicine refills.
- **Analytics Dashboard:** Access an overall admin dashboard showing all patients and doctors registered in the system.

Machine Learning Technologies

1. Smart Behavioural Feature Extraction Pipeline

- **Functionality:** Machine learning pipelines analyse dynamic patient behavioural data such as daily adherence checkmarks, medication refill latencies, etc. to discover complex hidden trends.
- **Benefits:** Translates raw data inputs into structured feature vectors, highlighting sudden behavioural dropouts before the patient health declines.

2. SMOTE (Synthetic Minority Over-sampling Technique)

- **Functionality:** An advanced data balancing framework that scientifically generates synthetic data profiles for high-risk, non-compliant patients to fix severe class imbalance issues within medical datasets.
- **Benefits:** Ensures that the predictive engine remains unbiased, heavily improving the AI model's precision and recall when identifying minority high-risk patients.

3. Predictive Analytics for Patient Dropout Mitigation

- **Functionality:** Supervised predictive models (specifically optimized **XGBoost and Random Forest** algorithms) run in the background to calculate a live **Adherence Risk Score (0-100%)** for each active patient.
- **Benefits:** Proactively flags unstable or non-adherent patient profiles on the clinician's dashboard, triggering early warnings so medical teams can step in with timely action before a health crisis occurs.

4. Voice Activated Medication Logging

- **Functionality:** Implements a lightweight, browser-native speech-to-text engine using the **Web Speech API**. This allows elderly or visually impaired patients to log their medication adherence dynamically using simple vocal commands.
- **Benefits:** Drastically lowers the barrier to entry for geriatric patients, ensuring consistent and high-quality adherence logging without manual interface navigation.

5. Prescription OCR Engine

- **Functionality:** Integrates an Optical Character Recognition (OCR) pipeline via **Tesseract OCR** inside the FastAPI microservice. Patients can simply upload an image of their physical doctor's prescription, and the engine automatically extracts the drug names and dosages to populate their digital pill checklist.
- **Benefits:** Eliminates manual data entry errors and streamlines the initial user onboarding process.

By integrating these innovative features and leveraging machine learning technologies, the "MediSense Non- Adherence Mitigation Portal" sets a new standard in medical arena, providing methods to prevent future medical emergencies at the earliest.

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